

YANG, Yan

<https://scholar.google.com/citations?user=LzdxmNcAAAAJ>

Degrees

Aug 2017 – May 2019 M.Sc. Georgetown University (Washington, DC)

Algorithms | Computer Hardware & System Architecture | Comp. Corpus Linguistics | Empirical Methods in NLP | Stat. Machine Translation | Adv Semantic Representation | Automatic Reasoning | Machine Learning | Deep Reinforcement Learning | Independent Study: Advanced Analytics | Statistical Machine Learning (Ph.D. seminar, approved by instructor) Supervisor: Prof. Grace Hui Yang

Sep 2002 – Jun 2006 B.Eng. Southeast University (Nanjing, China)

** (2003) Course Scholarship for Linear Algebra and Space Analytic Geometry*

Linear Algebra & Space Analytic Geometry (scholarship endowed) | Advanced Mathematics | Probability & Statistics | Numerical Computing & Modeling | Digital Image Processing | Speech Signal Processing | Signals & Linear System | Electromagnetic Field & Waves | Digital Systems & Course Design | Electronic Circuits & Comprehensive Experiment | Microcomputer Systems & Interfaces | Fundamentals of Circuit | Analog Electronic Circuits | Modern Psychology | Modern Biology | College Physics | Music Theory | Practice of MATLAB | etc.

Other education and expertise

(Mar 2026 - now) *The Principles of Quantum Mechanics*, P.A.M. Dirac (Ch 1; finishing Ch 2) self-study

(Oct 2025) *Autumn School on fMRI (EEG module cancelled)*, IBRO & New York University, Abu Dhabi

- Exposure to experimental design and analysis workflows (Python-based), fMRI principles (HRF, preprocessing), and notebook-based modelling pipelines (e.g., GLM and second-order models).

(Nov 2024 – May 2025) *Elements of Information Theory*, Tomas Cover (fully grasped Ch 1-8 and 14; touched on Ch 11, 13, and 15) self-study

(Aug – Sep 2024) MIT 18.102 *Introduction to Functional Analysis* (went over Lectures 1-8 and 14 Basic Hilbert Space Theory), self-study

(Sep 2023 - Nov 2024) Ph.D. Fellow, at KAUST (Thuwal, Saudi Arabia)¹

(2023) *Probabilistic Graphical Model 2* (94/100) & *Probabilistic Graphical Model 1* (100/100) on Coursera (Stanford University online)

(2022) *Computational Neuroscience* (99.57/100) on Coursera (University of Washington)

¹ *Departure was due to supervisory misconduct and institutional failures, both formally documented. The supervisor's published work is currently under journal-level review for research integrity concerns (<https://pubpeer.com/publications/7893CE90BA51FEEDCEF83F23B88FFA>). My assigned research direction was built upon conclusions that could not be verified—both those in the supervisor's published work and some proposed in subsequent supervisory discussions—and supervisors declined to engage with concerns, which obstructed the development of research output. I maintained independent research activity: ideas developed during this period, along with subsequent independent work, are documented on Zenodo, and I used the time to strengthen my theoretical foundations in mathematics and physics.*

(2022) *Human Research (Social Behaviour Research Investigators and Key Personnel Group, Basic)* – CITI Program & University of Nevada, Reno

(Jan 2021 – Aug 2022) Ph.D. studies, at University of Nevada, Reno (all courses done)

(2021) *Convolutional Neural Networks* (100/100) on Coursera (DeepLearning.AI)

(2021) *Mathematical Biostatistics Boot Camp I* (100/100) on Coursera (Johns Hopkins University)

(Spring 2021) Training: *GRAD 701S – College Teaching* @ University of Nevada, Reno

(2020) *Certified Generalist Software Engineer* (3% interviewees) – TripleByte

(2020) *Genomics: Decoding the Universal Language of Life* (unlicensed) on Coursera (UIUC)

(2020) *Finding Hidden Messages in DNA* (unlicensed) on Coursera (UC San Diego)

(2017) *Gender Study* - Harvard Summer School (writing A+; other: A) in Beijing (by Harvard University Director of Undergraduate Studies, Prof. Caroline Light)

Research output

Non-refereed and peer-reviewed scientific articles:

<Work related to cognitive motivational state modelling>

- Yang, Y. (2025). From Satiation to Allostasis: Toward a Resource-Based Framework for Reversal Triggers in Reversal Theory. *Zenodo*. <https://zenodo.org/records/17713710>
- Yang, Y. (2025). Modeling Limits in Phenomenological Contexts – a Statement on Improving Searcher Struggle Detection via the Reversal Theory. *ResearchGate Preprint*. [dx.doi.org/10.13140/RG.2.2.28606.80961](https://doi.org/10.13140/RG.2.2.28606.80961)
- Jiyun, L., Yang, Y., Nayak, V., & Yang, G.H. (2024). Improving Searcher Struggle Detection via the Reversal Theory. *Discover Computing* 27, 51. Springer Nature. doi.org/10.1007/s10791-024-09492-z

<Work related to generative modelling>

- Yang, Y. (2025). Towards PDE-Structured Generative Modeling: Differentiable Simulators, Flow-Matching, and Pattern Manifolds (Proposal). *Zenodo*. <https://doi.org/10.5281/zenodo.17897116>
- Yang, Y. (2026). Expectation-Maximization Style Algorithm for Task-Driven Differentiable Renderer Optimization. *Zenodo*. <https://zenodo.org/records/18666830>
- Yang, Y., Bebis, G., & Nicolescu, M. (2022). Two-Step Data Augmentation for Masked Face Detection and Recognition: Turning Fake Masks to Real. In *Proceedings of the 2nd International Conference on Image Processing and Vision Engineering – IMPROVE* (pp. 126-134). SciTePress. [dx.doi.org/10.5220/0011037900003209](https://doi.org/10.5220/0011037900003209)

Methods, software, infrastructures, materials, guides and tools developed

(2025) Investigated the feasibility of **gradient-based parameter learning** in **reaction–diffusion systems** (Gray–Scott model), focusing on **backpropagation** via **time-stepped PDE simulations**. After repeated failure modes, developed a structured **experimental codebase** that **enables systematic probing** of gradient behavior, numerical mechanisms, and optimizer dynamics via truncated BPTT, adaptive learning rates, and embedded simulation visualization. This work supports iterative refinement **toward a viable differentiable implementation**. Code is at: <https://github.com/Yan-Yang-bot/bp2renderer>. *Independent Research @ self*

(2021 ~ 2022) Led the prototyping of the Neural Network **embedding visualization system** and took charge of the **system design** of its remodelling, developing it into a **key “human-in-the-loop” step** of the Cluster-Based Sorting product. *R&D Intern / R&D Engineer @ Deepcell Inc.*

(2021) Developed **installation guides** for the local-running prototype version of CNN embedding visualization tool. *R&D Intern @ Deepcell, Inc.*

(2018) Revamped search engine's **VR interface (hand model formulation** and connection) for **behavioral data collection** and provided **web technique** support for its SIGIR conference demonstration, work acknowledged by the corresponding author as authorship-worthy. *Research Assistant @ Georgetown University*

(2015) Led the development of a weather report app with **dynamic charts** for precipitation and other weather metrics (AngularJS, Flask, NodeJS, AliCloud, Nginx, MySQL). **Data API** monitoring. **Precipitation interpolation.** *Full Stack Engineer @ ColorfulClouds Tech Co. Ltd*

Scientific and societal impact

(2025) Active Participant @ Models of Consciousness Conference 6

Contributed to meta-discussions on consciousness modeling by introducing **Gödel's incompleteness theorem**.

(2025) LinkedIn content creator

Visualisations & essays for **mathematical** understanding, audiences included academic professors, researchers, startup co-founders and top company engineers:

- **Animation:** Dirac delta approximation via cosine summation
- Visual diagram of matrix classifications in **advanced linear algebra**
- Short essay on the development of **non-Euclidean geometry**, with a visual timeline

(2022) Active Participant @ Models of Consciousness Conference 3

Raised discussion topics on **timescales of psychological phenomena** and related **data collection**; engaged in post-talk exchanges with faculty from CMU (CS, **subconscious models**) and Stanford (philosophy of mind, **qualia**).

(2022) Presenter @ IMPROVE (online due to COVID-19)

First-author paper *Two-Step Data Augmentation for Masked Face Detection and Recognition*.

Language skills

Mandarin – Native. **English** – Fluent C2.

6 years in the US; 7 years science/tech work in English; Graduate schools in English; Translated 19th century Old English in a full-length manuscript.

Certificates: Dec 2022, *Duolingo* 135; Nov 2016, *TOEFL* 105.

French – Basic A1.

Greetings; Recognize simple words, sentence structures, basic tenses in general books/packages/billboards.

Traditional Chinese – Basic A2.

Basic Duolingo & YouTube courses done; Simple talk with natives; Read beyond its Mandarin writing similarity.

Chronological work experience²

2025 Reviewer @ IEEE Transactions on Affective Computing

Reviewed machine-learning manuscripts:

- Identified deeper analytical questions necessary for interpreting empirical results.
- Corrected a serious conceptual misuse of cross-disciplinary terminology.
- Helped strengthen the coverage and selection of related work.
- Proposed structural revisions to the writing.

Oct 2025 – Current Independent Researcher

- Independently continued KAUST-era research on reaction-diffusion system parameter learning; developed a code repository with comprehensive visualization and logging tools for dynamical system parameter spaces and loss landscape analysis; experimented with direct backpropagation through unrolled Gray-Scott time steps, adaptive learning rates, and alternative loss functions; evaluated Intermediate Supervision, PINN, and surrogate model approaches as next steps.
- Participated in Models of Consciousness Conference 6 (Japan) and IBRO Autumn School on fMRI (NYU Abu Dhabi; EEG module cancelled).

Nov 2024 – Sep 2025 <Career Gap> *Freelance work, theoretical consolidation, and limited independent research; activities included journal reviewing, independent writing, mathematical visualization works, theoretical consolidation, and doctoral application preparation; included a medical recovery phase.*

2024 Reviewer @ IEEE Transactions on Neural Networks and Learning Systems

Sep 2023 – Nov 2024 Research Fellow @ KAUST (Thuwal, Saudi Arabia)

- Explored methods to learn reaction-diffusion systems (i.e., PDE parameters) from data; the underlying mathematics connects to neuroscience modeling, and in this project was applied to automate synthesis of diseased leaf images for agricultural classification models.
- Proposed ideas on Arabic Speech Recognition, including acoustic source separation and data augmentation in the spectrogram.

May – Aug 2023 <Career Gap> *Road trip across half the US and personal life-planning milestone after a layoff and concurrently obtaining a Ph.D. offer.*

2022 – 2023 Machine Learning R&D Engineer @ Deepcell Inc. (Mountain View, United States)

- Led the system design of the remodeled Neural Network embedding visualization system, which is to be used seamlessly in the Cluster-Based Sorting product. When the imaging phase of an imaging+sorting run is done, we use this design to pause the run and start a data exploration session to select sorting criteria.
- My system design covered decisions on data storage media, database schemas, API specs, serialization/deserialization options, caching, parallelizing large calculations, and others for incremental developing plans and long-term goals.

Sep 2022 <Career Gap> *One-month gap between OPT/H-1B work permits (not allowed to work by the US Immigration Bureau); attended Models of Consciousness Conference 3.*

² This section outlines all roles, academic, industrial, and volunteer, intentionally arranged chronologically to reflect the evolution of my research and professional interests

2022 Summer Volunteer Research Assistant @ Visual Perception Lab - Neuroscience, University of Nevada, Reno (Reno, United States)

- Trained for BrainVoyager usage and human research ethics.
- Helped with software and package installations and documented an installation guide.
- Read papers on Representational Similarity Analysis, Multivoxel Pattern Analysis, human brain's face/object perception, fMRI, etc., & discussed research ideas with other lab members based on their existing code and data.

2022 Spring Teaching Assistant @ University of Nevada, Reno (Reno, United States)

Recitation classes, grading, office hours, & exam proctorship, MATH 127, Precalculus II, including preparation of structured recitation slides and class exercises.

2021 – 2022 R&D Intern @ Deepcell Inc. (Mountain View, United States)

- Trained GAN models, supplementing data for cell image classifiers.
- Led the development of a comprehensive CNN embedding layer visualization tool, used company-wide by data and bioinformatic scientists for clustering, correlation analysis, etc., and greatly improved their productivity.

2021 Researcher @ University of Nevada, Reno (Reno, United States)

Published Two-Step Data Augmentation for Masked Face Detection and Recognition: Turning Fake Masks to Real.³

2021 Teaching & Research Assistant @ University of Nevada, Reno (Reno, United States)⁴

Grading and office hours, CS 320, Data Structure (C++).

2020 – 2021 Member of Technical Staff @ Pure Storage (Bellevue, United States)

- Implemented feature: import/restore volume snapshots from company disks into Kubernetes.
- Enhancement/bug fixing: multipath device problem, upgrading bash scripts to Python3, etc.
- Led the GitHub release process of Pure Service Orchestrator v 6.0.3.

2019 – 2020 <Career Gap> Completed research postponed due to making up coursework after an in-semester medical leave; began job search, with placement delayed by the onset of COVID-19.

2018 – 2019 Research Assistant @ Georgetown University (Washington, United States)⁵

³ Completed as a coursework project under significant constraints: self-funded through a concurrent internship, new topic initiated in the second week due to privacy restrictions, partial self-collected data, and delayed compute access. The original coursework writeup was submitted without modification and accepted without revision requirements. It was subsequently invited for revision for top conference submission, but further submission was not pursued due to resource limitations.

⁴ Research assistant funding was cut abruptly; subsequently pursued an industry internship and a teaching role to bridge the gap while continuing to seek research opportunities.

⁵ To avoid confusion, during this period, I contributed to two research projects: one resulting in a journal publication five years later (2024), and another involving a VR interface for SIGIR 2018, which the corresponding author recognized as authorship-worthy.

@InfoSense Lab: 1) Translated psychological reversal theory into statistical languages to analyze search engine user behaviors (search logs) and improve user struggle detection; 2) Revamped the search engine's VR interface for behavioral data collection, and provided web technique support for its SIGIR '18 conference demonstration, work acknowledged by the corresponding author as authorship-worthy.

@Massive Data Institute: "Victims of Violence" tweets analytics: occupational analysis and database management.

2016 – 2017 <Career Gap> Language training and graduate-school applications; attended Harvard Summer School (Gender Study).

2016 Web Development Engineer & Mentor @ DataMesh (Beijing, China)

- Led the development of a news publishing system implemented as a React-Redux app;
- Solved Apache deployment conflicts among static files, React router hack, and Python service;
- Mentored new employees on webpage building.

2016 Software Engineer @ Delicious Meals Delivery Tech Ltd. (Beijing, China)

Designed and implemented a message push system using WebSocket, AngularJS, and GoLang Micro Service for the meal website, which solved user complaints that pages don't sync the latest meal status.

2015 – 2016 Full Stack Engineer @ ColorfulClouds Tech Ltd. (Beijing, China)

- Independently took charge of a weather report app with dynamic charts for weather metrics (e.g., precipitation).
- Implemented a compatibility patch for data API v2, providing continued support to v1 clients.
- Automatic periodic collection of API health data into the Redis database.
- Precipitation interpolation using NumPy.

2014 – 2015 Frontend Developer @ One Lead Interactive Media Advertising Co., Ltd., etc. (Beijing, China)

- Made WeChat minigames for marketing activities: <https://github.com/yan-yang-bot/MiniGame-Zillionaire>.
- Built webpages for an information management app, using D3.js for data reports and charts.

2013 – 2014 Web Development Engineer @ EnglishPass100 Education Tech. Ltd., etc. (Beijing, China)

Developed C++-based Windows apps (ID card verifier, Word Solitaire game) with Linux server.

2013 <Career Gap> Career transition from translation to software engineering via **self-study in C++**; **independently developed a minesweeper game.**

2009 – 2012 Translator (EN-CN) @ Self-employed (Beijing, China)

- Translated and organized the book *Women in the 19th Century* (Margaret Fuller, 1843), ensuring accuracy and consistency across a full-length manuscript.
- Translated and proofread marketing materials, user/developer manuals, TV subtitles, and software/game content for clients including Yeeyan.org and Grand Strong.

<Scientific News Translator> from 2009 to 2010

- Published dozens of Chinese translations of science and technology news from media outlets such as Wired.com.

2008 – 2009 Scientific Secretary @ Tongji University (Shanghai, China)

2006 – 2008 <Career Gap> *Mental health recovery following sexual misconduct and other incidents in undergraduate; engaged in **self-study of psychology and philosophy**. (Details omitted for academic focus.)*

2006 – 2008 Psychology Peer Supporter (Volunteer) @ Ayaohelp.com (Remote, China)

- Provided **peer support** informed by **self-study in psychology and philosophy**, focusing on strategic life-event analysis and knowledge sharing. In addition to practice-oriented interpretive approaches commonly used in counselling contexts, I intentionally covered topics in experimental psychology and general psychology, e.g., cognition, perception, learning, memory, basic principles of research design, measurement, and validity;
- Reflective **essays/poems** on **personal development** were **highlighted on front page** by the site host, a Psychology professor at Beijing Normal University, to engage readers.